The Radiance Equation

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Outline

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Introduction

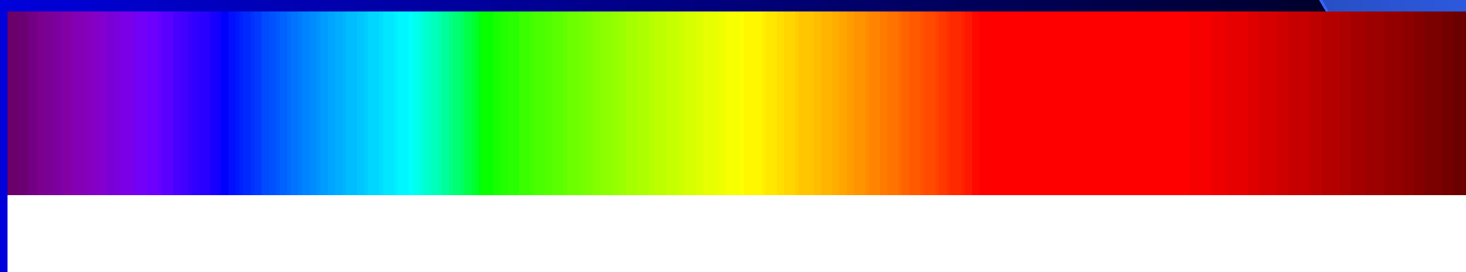
- Lighting is the central problem of real-time graphics rendering
  - Arbitrary shaped lights
  - Changes in lighting conditions
  - Real-time shadows
  - Real-time reflections
  - Mixtures of many different types of surface

Introduction

- Real-time walkthrough with global illumination
  - Possible under limited conditions
    - Radiosity (diffuse surfaces only)
- Real-time interaction
  - Not possible except for special case local illumination
- Why is the problem so hard?

Light

- Visible light is electromagnetic radiation with wavelengths approximately in the range from 400nm to 700nm



400nm 700nm

Light: Photons

- Light can be viewed as wave or particle phenomenon
- Particles are photons
  - packets of energy which travel in a straight line in vacuum with velocity *c* (300,000 km.p.s.)
- The problem of how light interacts with surfaces in a volume of space is an example of a *transport* problem.

Light: Radiant Power

- Φ denotes the *radiant energy* or *flux* in a volume V.
- The flux is the rate of energy flowing through a surface per unit time (watts).
- The energy is proportional to the particle flow, since each photon carries energy.
- The flux may be thought of as the flow of photons per unit time.

Light: Flux Equilibrium

- Total flux in a volume in dynamic equilibrium
  - Particles are flowing
  - Distribution is constant
- Conservation of energy
  - Total energy input into the volume = total energy that is output by or absorbed by matter within the volume.

Light: Fundamental Equation

- Input
  - Emission – emitted from within volume
  - Inscattering – flows from outside
- Output
  - Streaming – without interaction with matter in the volume
  - Outscattering – reflected out from matter
  - Absorption – by matter within the volume
- Input = Output

Light: Equation

- $\Phi(p,\omega)$ denotes flux at $p \in V$, in direction ω
- It is possible to write down an integral equation for $\Phi(p,\omega)$ based on:
  - Emission+Inscattering = Streaming+Outscattering + Absorption
- Complete knowledge of $\Phi(p,\omega)$ provides a complete solution to the graphics rendering problem.
- Rendering is about solving for $\Phi(p,\omega)$.

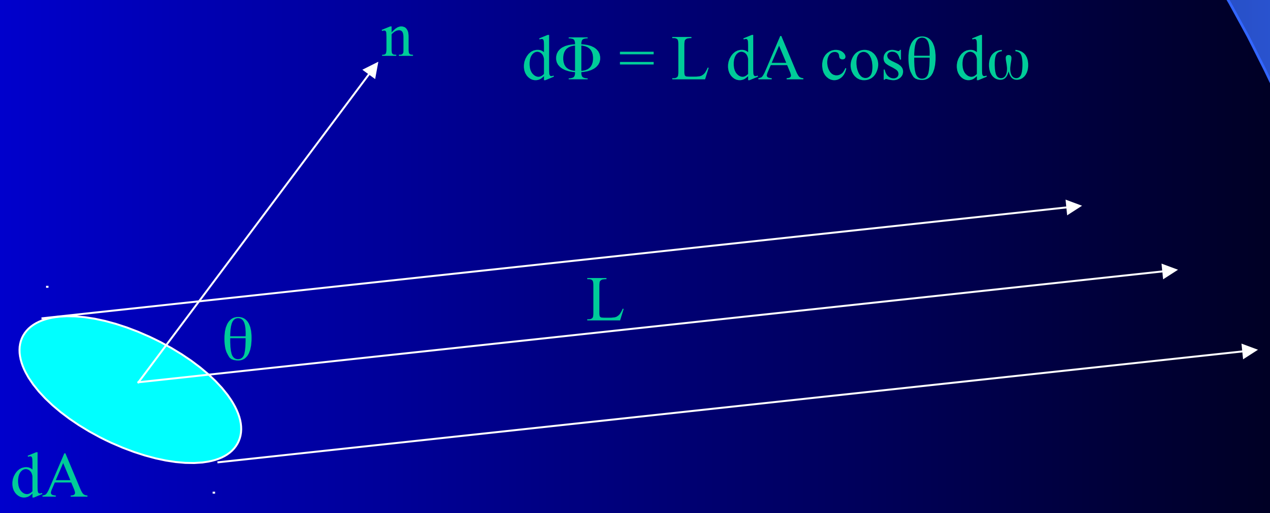
Simplifying Assumptions

- Wavelength independence
  - No interaction between wavelengths (no *fluorescence*)
- Time invariance
  - Solution remains valid over time unless scene changes (no *phosphorescence*)
- Light transports in a vacuum (non-participating medium) –
  - ‘free space’ – interaction only occurs at the surfaces of objects

Radiance

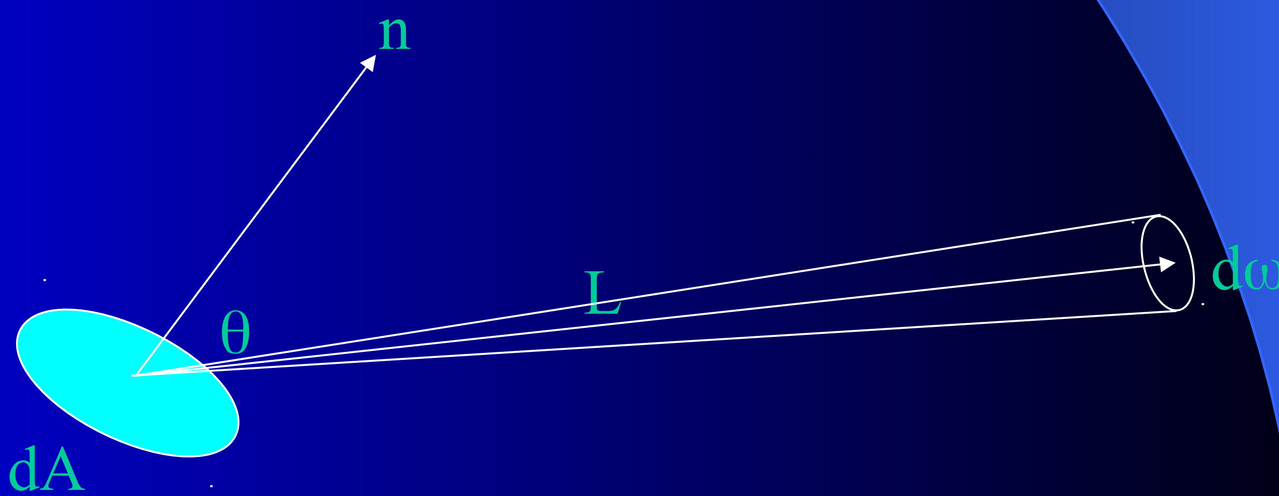
- Radiance (L) is the flux that leaves a surface, per unit projected area of the surface, per unit solid angle of direction.

$$d\Phi = L \, dA \cos\theta \, d\omega$$



Radiance

- For computer graphics the basic particle is not the photon and the energy it carries but the ray and its associated radiance.



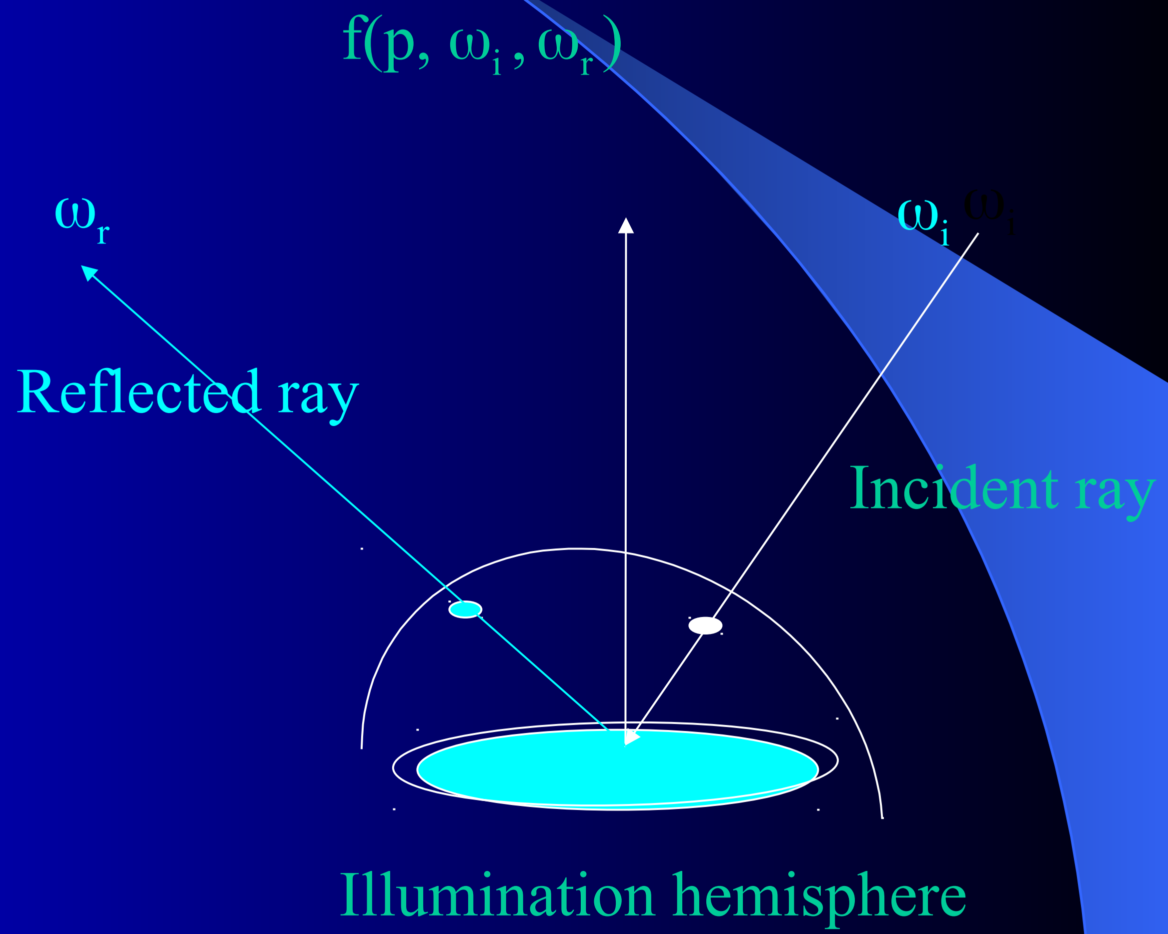
Radiance is constant along a ray.

Radiance: Radiosity, Irradiance

- Radiosity - is the flux per unit area that radiates from a surface, denoted by *B*.
  - $d\Phi = B\,dA$
- Irradiance is the flux per unit area that arrives at a surface, denoted by *E*.
  - $d\Phi = E\,dA$

Reflectance

- BRDF
  - Bi-directional
  - Reflectance
  - Distribution
  - Function
- Relates
  - Reflected radiance to incoming irradiance



Reflectance: BRDF

- Reflected Radiance = BRDF$\times$Irradiance
- $L(p, \omega_r) \quad = f(p, \omega_i, \omega_r)\, E(p, \omega_i)$
- $\quad = f(p, \omega_i, \omega_r)\, L(p, \omega_i) \cos\theta_i \, d\omega_i$

- In practice BRDF's hard to specify
- Rely on ideal types
  - Perfectly diffuse reflection
  - Perfectly specular reflection
  - Glossy reflection
- BRDFs taken as additive mixture of these

The Radiance Equation

- Radiance $L(p, \omega)$ at a point p in direction ω is the sum of
  – Emitted radiance $L_e(p, \omega)$
  – Total reflected radiance

Radiance = Emitted Radiance + Total Reflected Radiance

The Radiance Equation: Reflection

- Total reflected radiance in direction ω:
  - $\int f(p, \omega_i, \omega)\, L(p, \omega_i) \cos\theta_i \, d\omega_i$
- Radiance Equation:
- $L(p, \omega) = L_e(p, \omega) + \int f(p, \omega_i, \omega)\, L(p, \omega_i) \cos\theta_i \, d\omega_i$
  - (Integration over the illumination hemisphere)

The Radiance Equation

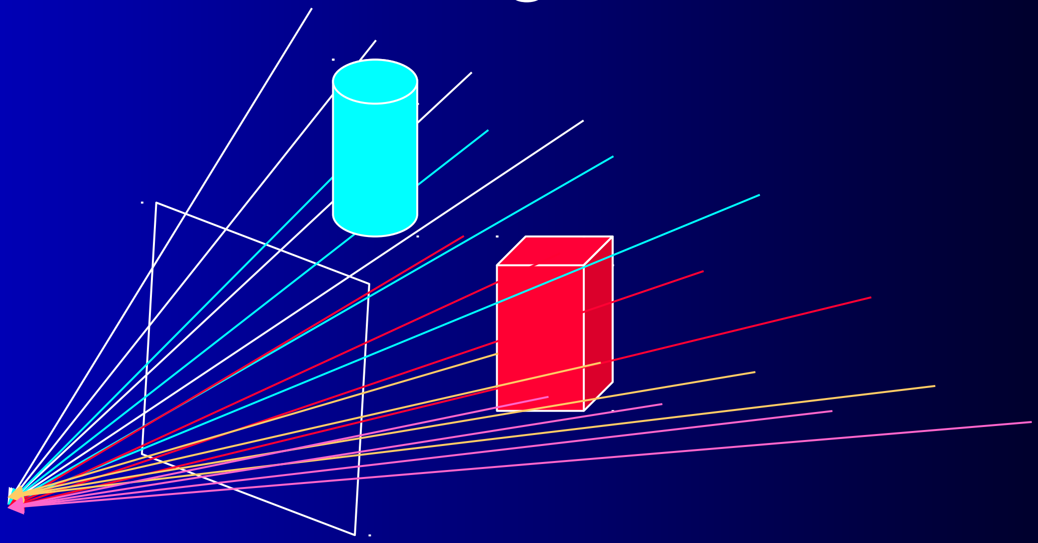
- p is considered to be on a surface, but can be anywhere, since radiance is constant along a ray, trace back until surface is reached at p*, then
  - $L(p, \omega_i) = L(p^*, \omega_i)$

$L(p, \omega)$ depends on all $L(p^*, \omega_i)$ which in turn are recursively defined.

The radiance equation models global illumination.

Traditional Solutions to the Radiance Equation

- The radiance equation embodies totality of all 2D projections (view).
- Extraction of a 2D projection to form an image is called rendering.



Traditional Solutions

	Local Illumination	Global Illumination
View Dependent	‘Real time’ graphics: OpenGL	Ray tracing Path tracing
View Independent	Flat shaded graphics (IBR)	Radiosity (Photon Tracing)

(Image Based Rendering)

- IBR not a ‘traditional’ solution
- Images for a new view constructed from a large collection of existing images
- No lighting computations at all.
- Light Field Rendering specific instance to be discussed later.

Visibililty

- Where does an incident ray through the image plane come from?
  - Which surface?
- Ray tracing in principle has to search all surfaces for possible intersections
- Radiosity has to include visibility in form-factor calculations between surfaces
- Real-time rendering solves visibility problem on a pixel by pixel basis (z-buffer).
  - Major complication for large scenes
- We will see later that LFR does not have this visibility problem.

Conclusions

- Graphics rendering is concerned with solution of integral radiance equation
- Traditional solutions are various kinds of approximations to this equation.
- Rendering is the process of extracting images from the equation.
- Rendering may be view dependent or independent, together with a global or local illumination solution.